

Jingzhe Guo

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EDUCATION

Tsinghua University

PH.D. IN COMPUTER SCIENCE AND TECHNOLOGY

Cumulative GPA: **3.94** / 4.00

Sep. 2020 -

Tsinghua University

B.S. IN COMPUTER SCIENCE AND TECHNOLOGY

Cumulative GPA: **3.73** / 4.00

Aug. 2016 - Jun. 2020

RESEARCH INTEREST

- Compilation of quantum programs.
- Design of quantum programming languages.

RESEARCH

isQ, Quantum programming language and quantum software stack

- Participated in **design** of isQ quantum programming language and isQ-IR, an MLIR-based intermediate representation for transformation of quantum programs.
- Participated in **implementation** of the isQ software stack, including front-end, MLIR-based compiler middle-end, real-hardware code generator and quantum simulator back-end.

PUBLICATIONS AND PRESENTATIONS

Publications:

- **Jingzhe Guo**, Mingsheng Ying: *Software Pipelining for Quantum Loop Programs*. IEEE Trans. Software Eng. 49(4): 2815-2828 (2023)
- **Jingzhe Guo**, Huazhe Lou, Jintao Yu, Riling Li, Wang Fang, Junyi Liu, Peixun Long, Sheng-gang Ying, Mingsheng Ying. *isQ: An Integrated Software Stack for Quantum Programming*. IEEE Transactions on Quantum Engineering, vol. 4, pp. 1-16, 2023, Art no. 2500116, doi: 10.1109/TQE.2023.3275868.

Presentations:

- *isQ: An Integrated Software Stack for Quantum Programming*. 5th International Workshop on Quantum Compilation, 22-23 July 2023, Paris, France.

SKILLS

Programming Languages Rust, Haskell, Typescript, C, C++, Python, Golang

Tools MLIR, Nix